

8(a)(i)	Y and Z have equal numbers of protons and (so) they have the same charge	B1
8(a)(ii)	Y has (two) fewer protons (than X)	M1
	(so) Y has less charge (than X)	A1
8(b)	(total) momentum before decay is zero or X has zero / no momentum	B1
	(total momentum after decay must be zero so) Y must have equal (and opposite) momentum to α -particle (so cannot be stationary / must have speed/velocity)	B1